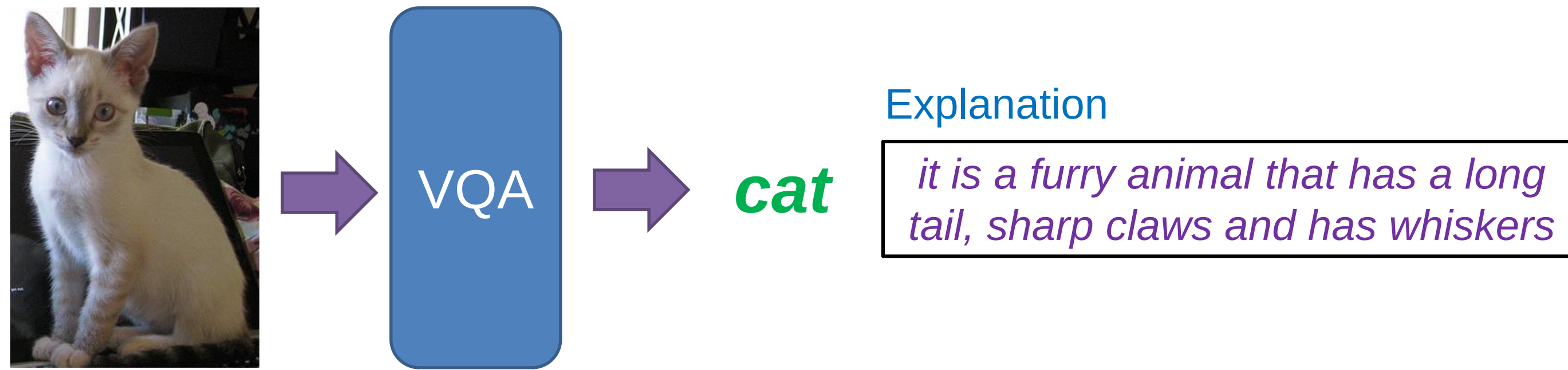


Natural Language Explanations

Stepping away from heatmaps

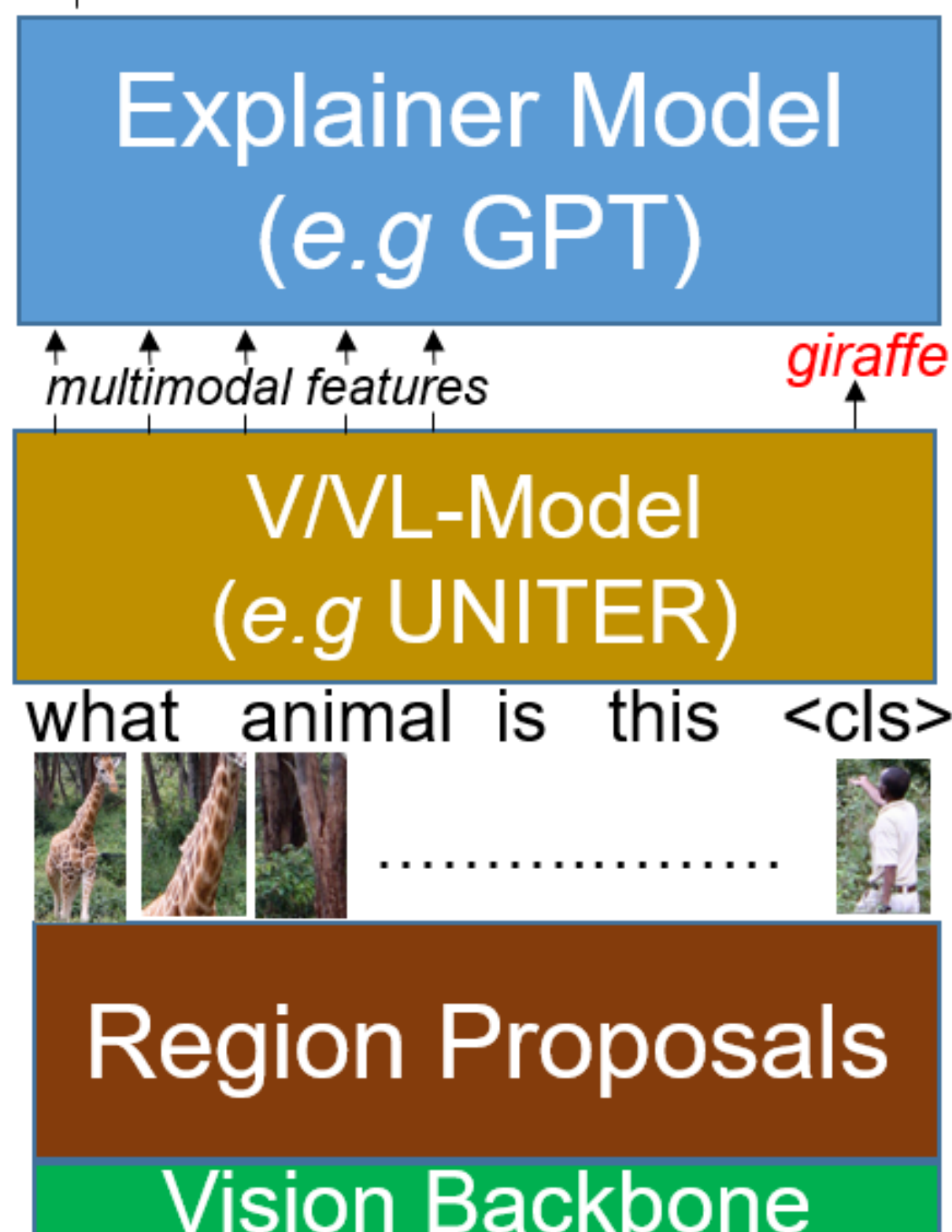


What animal is this?

Problems in Current Models

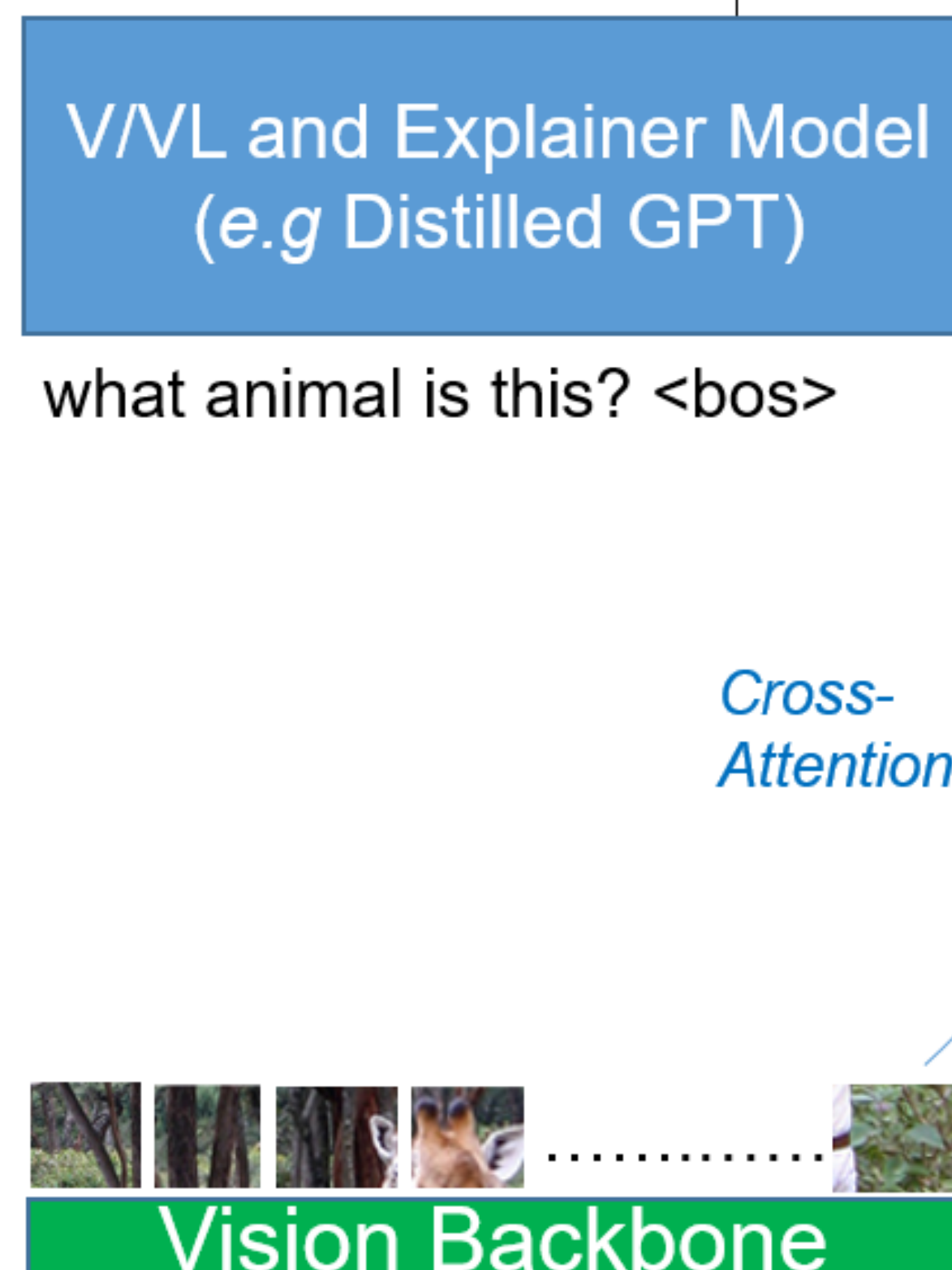
- 1) Requires a Vision or VL-model to predict the answer
- 2) The Vision or VL-model and the Explanation Model are completely independent pipelines. The same model which gave the answer should also be used to give the explanation

it has a long neck and black spots



(a) Previous Models

giraffe because it has...

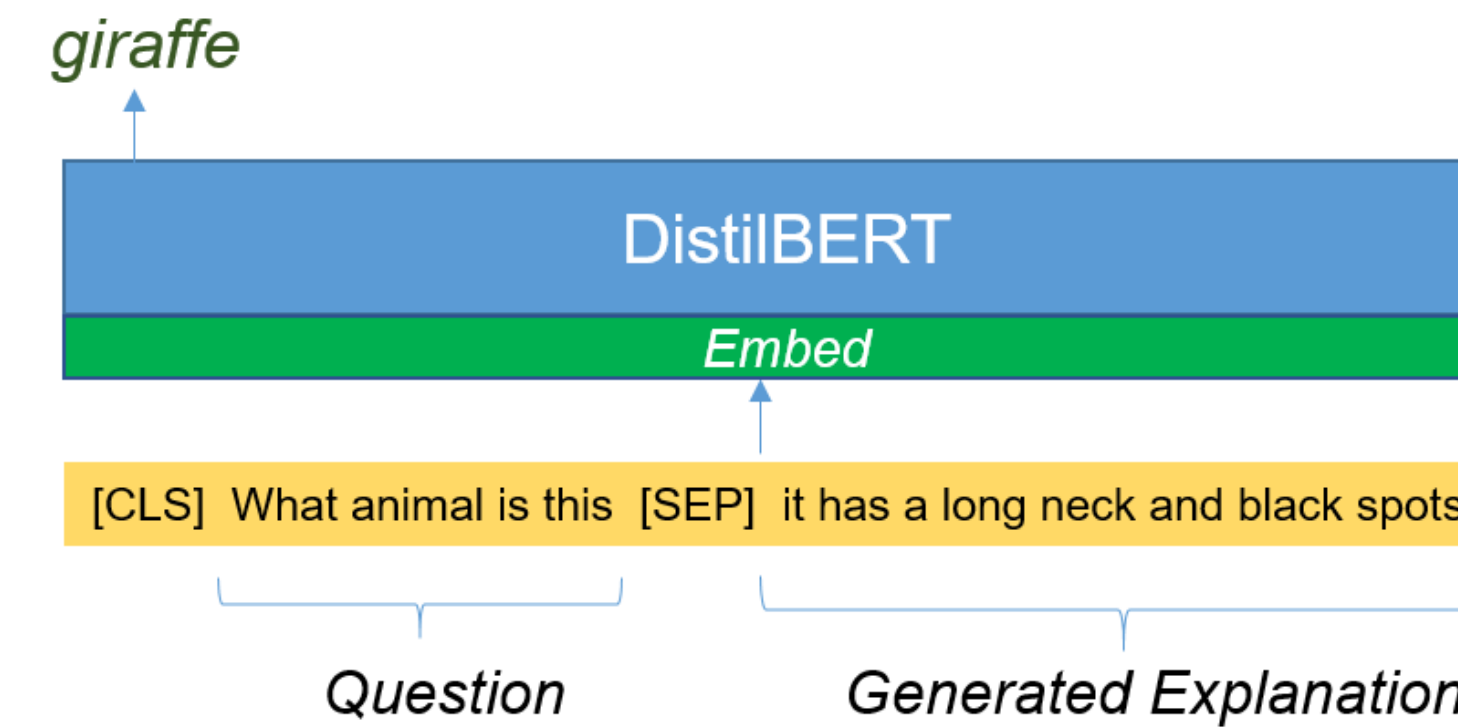


(b) Our Model

Evaluation Metrics

We propose two evaluation metrics:

a) Retrieval-based Attack and b) Explain-Predict



Quantitative Evaluation

Approach	VQA-X					ACT-X					
	B4	M	R	C	S	B4	M	R	C	S	Human
CAPS [40]	5.9	12.6	26.3	35.2	11.9	5.2	11.0	26.5	10.4	4.6	22.9
PJ-X [40]	19.5	18.2	43.4	71.3	15.1	15.3	15.6	40.0	22.0	7.2	38.2
FME [60]	24.4	19.5	47.4	88.8	17.9	-	-	-	-	-	-
NLX-GPT (w/o pretraining)	23.8	20.3	47.2	89.2	18.3	25.6	21.4	48.0	63.5	15.4	-
NLX-GPT (w/ pretraining)	25.6	21.5	48.7	97.2	20.2	28.1	22.6	49.7	74.9	17.6	89.0

VQA-X											
	B1	B2	B3	B4	R-L	M	C	S	BS	Human	Task Acc.
PJ-X [40]	57.4	42.4	30.9	22.7	46.0	19.7	82.7	17.1	84.6	65.4	76.4
FME [60]	59.1	43.4	31.7	23.1	47.1	20.4	87.0	18.4	85.2	63.2	75.5
RVT [35]	51.9	37.0	25.6	17.4	42.1	19.2	52.5	15.8	85.7	67.1	68.6
QA-only [22]	51.0	36.4	25.3	17.3	41.9	18.6	49.9	14.9	85.3	-	-
e-UG [22]	57.3	42.7	31.4	23.2	45.7	22.1	74.1	20.1	87.0	71.5	80.5
NLX-GPT	64.2	49.5	37.6	28.5	51.5	23.1	110.6	22.1	86.9	83.22	83.07

e-SNLI-VE											
	B1	B2	B3	B4	R-L	M	C	S	BS	Human	Task Acc.
PJ-X [40]	29.4	18.0	11.3	7.3	28.6	14.7	72.5	24.3	79.1	59.6	69.2
FME [60]	30.6	19.2	12.4	8.2	29.9	15.6	83.6	26.8	79.7	58.5	73.7
RVT [35]	29.9	19.8	13.6	9.6	27.3	18.8	81.7	32.5	81.1	59.4	72.0
QA-only [22]	29.8	19.7	13.5	9.5	27.0	18.7	80.4	32.1	81.1	-	-
e-UG [22]	30.1	19.9	13.7	9.6	27.8	19.6	85.9	34.5	81.7	68.9	79.5
NLX-GPT (w/o Concepts)	35.7	24.0	16.8	11.9	33.4	18.1	114.7	32.1	80.6	66.3	-
NLX-GPT (w/ Concepts)	37.0	25.3	17.9	12.9	34.2	18.8	117.4	33.6	80.8	67.4	73.91

Qualitative Results



Is it raining?
yes because the ground is wet and people are using umbrellas



What beverage is in the cup?
coffee because it is a dark liquid with a dark brown top



What action is this?
skydiving because he is in the air with a parachute on his back and is falling off



What action is this?
cooking because she is standing in a kitchen and cutting up food on a cutting board



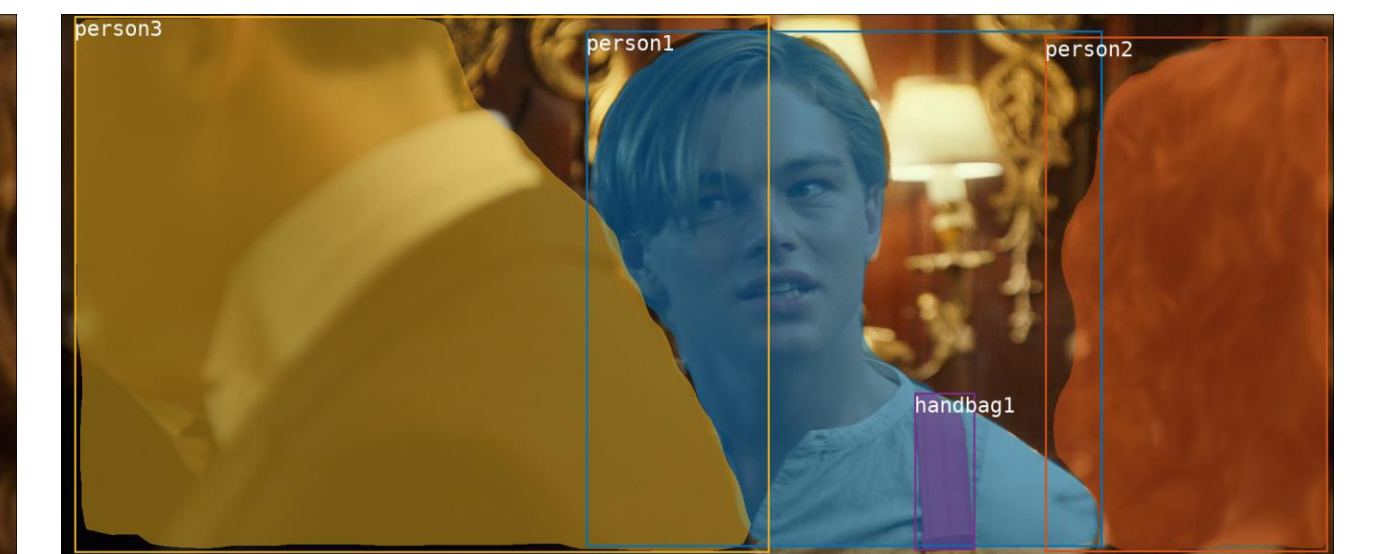
The animal is playing outside
entailment because a dog is an animal and a field is outside



A girl is sitting at home watching TV
contradiction because the girl cannot be sitting at home and playing on a playground simultaneously



who is person1 talking to?
person1 is talking to person3 because person1 is looking at person3 and is speaking to him



Code: <https://github.com/fawazsammani/nlxgpt>
Gradio Web-Demo: <https://huggingface.co/spaces/Fawaz/nlx-gpt>